Section A

Concept Application

➢ SCENARIO 1:

Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

❖ Question:

Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

1. CPU (Processor): Choose a processor with good speed and multiple cores so employees can run programming software, web browsers, and video calls at the same time.

2. RAM: Choose at least 16 GB RAM for smooth multitasking. Also, check that the RAM type and speed are supported by the motherboard.

3. Storage: Choose a 512 GB SSD because it loads programs and files faster than a traditional HDD.

4. Compatibility: Make sure the CPU matches the motherboard socket and supported chipset. Check that the RAM type is supported and the motherboard has the correct storage connection, such as M.2 NVMe or SATA.

5. Upgrade options: Choose a computer that allows RAM and storage to be upgraded in the future.

➢ SCENARIO 2:

After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

❖ Question:

Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

1. CPU: Antivirus software uses the processor to scan files and detect viruses. During a scan, the computer may become slower.

2. RAM: Antivirus software uses some RAM to run in the background. If many applications are open, less RAM is available for other programs.

3. Storage: Antivirus software scans files stored on the SSD or HDD. This can make opening files or loading programs slower during a scan.

4. System Slowdown: When antivirus software uses too much CPU, RAM, or storage resources, other applications may respond slowly.

➢ SCENARIO 3:

A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

❖ Question:

Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

1. Restart the Computer: Restart the computer to fix temporary problems.

2. Check Device Drivers: Open Device Manager and check whether any device has a warning sign or driver error.

3. Update Drivers: Download and install the correct drivers from the computer or hardware manufacturer's official website.

4. Update the Operating System: Check for Windows updates and install the required updates to fix bugs and improve performance.

5. Check System Files: Use the Command Prompt as administrator and run sfc /scannow to check and repair corrupted Windows system files.

6. Scan for Viruses: Run an antivirus scan to check whether malware is causing the slowdown.

7. Restart and Test: Restart the computer and check whether it is working properly.

➢ SCENARIO 4:

Nexus Innovations experiences frequent power cuts and voltage fluctuations.

❖ Question:

Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

A UPS (Uninterruptible Power Supply) and a surge protector help protect computer hardware from electrical problems and prevent data loss.

1. UPS: A UPS provides temporary battery power when the electricity goes off. It gives users time to save their work and shut down the computer safely.

2. Surge Protector: A surge protector protects computer components from sudden increases in voltage that can damage the motherboard, CPU, and other hardware.

3. Prevents Data Loss: If the power goes off suddenly, a UPS helps prevent unsaved work from being lost and reduces the risk of file corruption.

4. Protects Hardware: Both devices help reduce the risk of damage caused by electrical problems.

➢ SCENARIO 5:

New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

❖ Question:

Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

DNS and DHCP work together to help computers connect to the network and access websites.

1. DNS (Domain Name System): DNS converts website names, such as www.google.com, into IP addresses so the computer can find the correct server.

2. DHCP (Dynamic Host Configuration Protocol): DHCP automatically assigns an IP address, subnet mask, default gateway, and DNS server address to each computer on the network.

3. How They Work Together: DHCP gives the computer its network settings, including the DNS server address. DNS then helps the computer find websites using their names.

4. Which Service Is Misconfigured? The answer depends on the symptoms:

- If the computer cannot get an IP address or connect to the network, DHCP may be misconfigured.
- If the computer can access websites using IP addresses but not website names, DNS is likely misconfigured.

➢ SCENARIO 6:

After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

❖ Question:

Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

1. Check the Firewall Settings: Open the computer's firewall settings and check whether any rules are blocking the required application or network traffic.

2. Identify the Problem: Check the firewall logs to find blocked connections and identify the application, IP address, or port involved.

3. Test Network Connectivity: Use commands such as ping or Test-NetConnection to check whether the connection is working. A failed ping alone does not prove that the firewall is blocking traffic.

4. Review Firewall Rules: Check whether the required ports and protocols are allowed and whether any conflicting rules are present.

5. Test the Firewall Carefully: If necessary, test with a temporary, limited rule for the specific application or port. Avoid disabling the entire firewall.

6. Fix the Issue: Create or correct a firewall rule that allows legitimate traffic while keeping unnecessary connections blocked.

7. Verify the Solution: Test the application again, review the firewall logs, and confirm that the connection works correctly.

Section B

Practical Application

- Task 1:

Workstation Assembly

- Install CPU, RAM, storage, and other core components into the chassis in the correct sequence.

Steps to Install Computer Components

1. Prepare the Chassis: Turn off the power, unplug the computer, and place the chassis on a clean surface. Use an anti-static wrist strap to protect components from static electricity.

2. Install the CPU: Place the motherboard on a safe surface. Open the CPU socket, align the CPU with the marked corner, and gently place it into the socket. Lock the socket without forcing it.

3. Install the RAM: Open the RAM slot clips and align the notch on the RAM stick with the slot. Press gently until the clips lock into place.

4. Install the Storage: Insert the M.2 SSD into the motherboard slot and secure it with the screw. For a SATA SSD or HDD, mount it in the drive bay and connect the SATA data and power cables.

5. Install the CPU Cooler: Apply thermal paste if required, mount the CPU cooler, and connect its fan cable to the CPU_FAN header.

6. Install the Motherboard: Fit the motherboard into the chassis, align it with the standoffs, and secure it with screws.

7. Install the Power Supply (PSU): Secure the PSU in its designated area and connect the motherboard power cable and CPU power cable.

8. Connect Other Components: Install the graphics card if needed. Connect the case fans, front-panel buttons, USB ports, and other required cables.

9. Check All Connections: Ensure that the components are secure and the cables are connected correctly.

10. Start the Computer: Connect the monitor and power cable, switch on the computer, and enter the BIOS/UEFI to check whether the CPU, RAM, and storage are detected.

- ❖ Connect all required internal cables (power, data, front panel).

Steps to Connect Internal Computer Cables

1. Power Cables: Connect the 24-pin ATX power cable to the motherboard and the 4/8-pin CPU power cable near the processor. Connect the PSU power cables to the GPU and storage devices if required.

2. Data Cables: Connect the SATA data cable from the motherboard to the HDD or SATA SSD. M.2 NVMe SSDs connect directly to the motherboard and do not need a separate data cable.

3. Front-Panel Cables: Connect the power switch, reset switch, power LED, and HDD LED cables to the motherboard's front-panel header according to the motherboard manual.

4. Front USB and Audio: Connect the case's USB ports to the USB headers and the headphone/microphone port to the HD_AUDIO header.

5. Fan Cables: Connect the CPU cooler fan to the CPU_FAN header and case fans to the appropriate SYS_FAN or CHA_FAN headers.

6. Check Connections: Make sure all connectors are firmly seated and correctly aligned. Never force a connector into the wrong socket.

7. Test the Computer: Before switching on, check all connections. Then power on the computer and verify that the fans spin and the system starts normally.

- ❖ Document each installation step in order.

Computer Hardware Installation Report

Objective: To install the CPU, RAM, storage, and other core components into the computer chassis and connect all required cables.

Installation Steps in Order

1. Prepare the Chassis: Unplug the power supply, prepare a clean workspace, and use anti-static protection.

2. Install the CPU: Place the CPU into the motherboard socket, align it correctly, and lock it in place.

3. Install the RAM: Insert the RAM sticks into the recommended motherboard slots until they click into place.

4. Install Storage: Secure the SSD or HDD in its designated location. Connect the SATA data cable if required.

5. Install the CPU Cooler: Mount the cooler securely and connect its fan cable to the CPU_FAN header.

6. Install the Motherboard: Place the motherboard on the chassis standoffs and secure it with screws.

7. Install the Power Supply: Secure the PSU inside the chassis and prepare the required power cables.

8. Connect Power Cables: Connect the 24-pin motherboard power cable and the CPU power cable. Connect additional power cables for the GPU or storage if needed.

9. Connect Data Cables: Connect SATA data cables to SATA storage devices. M.2 SSDs do not require separate data cables.

10. Connect Front-Panel Cables: Connect the power switch, reset switch, power LED, and HDD LED according to the motherboard manual.

11. Connect Other Cables: Connect front USB, audio, and case-fan cables to their correct motherboard headers.

12. Final Inspection: Check that all components are secure and all cables are connected correctly.

13. Power On and Test: Connect the monitor and power cable, switch on the computer, and enter BIOS/UEFI to confirm that the CPU, RAM, and storage are detected.

Final Result

All core computer components were installed in the correct sequence, and the required internal cables were connected. The final BIOS/UEFI check confirms whether the hardware is detected and ready for operating system installation.

- ❖ System must pass POST (Power-On Self-Test) after assembly.

Steps to Perform POST

1. Check Connections: Ensure the CPU, RAM, storage, motherboard power cable, and CPU power cable are connected correctly.

2. Connect Peripherals: Connect the monitor, keyboard, and power cable.

3. Switch On the Computer: Turn on the power supply and press the computer's power button.

4. Observe the Display: Check whether the manufacturer's logo or BIOS/UEFI screen appears on the monitor.

5. Check BIOS/UEFI: Confirm that the CPU, RAM, and storage devices are detected correctly.

6. Check Error Indicators: Look for motherboard diagnostic LEDs, beep codes, or error messages that indicate hardware problems.

7. Troubleshoot if Needed: If POST fails, switch off and unplug the computer before checking connections or reseating components.

- Task 2:

RAM Mismatch Troubleshooting

- Identify the RAM mismatch error via BIOS/UEFI POST messages.

Identifying a RAM Mismatch Error via BIOS/UEFI POST

1. Start the Computer: Turn on the computer and observe the screen for any POST error messages.

2. Check the Error Message: Look for messages such as “Memory Error,” “Memory Configuration Error,” or “Memory Not Detected.”

3. Open BIOS/UEFI: Enter BIOS/UEFI and check whether the installed RAM capacity is detected correctly.

4. Check RAM Compatibility: Verify that the RAM type, speed, and capacity are supported by the motherboard.

5. Inspect RAM Installation: Turn off and unplug the computer, then check that the RAM sticks are properly seated in the recommended slots.

6. Test the RAM: If the error continues, test one RAM stick at a time to identify a faulty or incompatible module.

- ❖ Determine the root cause (mismatched speed, timing, or capacity).

1. Check RAM Speed: Open BIOS/UEFI and check the RAM speed. If the RAM speed is not supported by the motherboard, it may cause compatibility issues.

2. Check RAM Timing: Check the RAM timing settings. Incorrect settings or an unstable XMP/EXPO profile may prevent the computer from passing POST.

3. Check RAM Capacity: Verify the total RAM capacity and ensure that each RAM module is supported by the motherboard.

4. Test RAM Modules: Turn off and unplug the computer. Test each RAM stick individually to check whether one module is faulty or incompatible.

5. Check the Motherboard Manual: Confirm the supported RAM type, maximum capacity, speed, and recommended RAM slot configuration.

Root Cause: The problem may be caused by unsupported RAM speed, incorrect timing settings, incompatible module capacity, or a faulty RAM stick. Without the actual BIOS/UEFI error message and RAM specifications, the exact cause cannot be confirmed.

- ❖ Resolve the mismatch by reconfiguring or reseating the RAM correctly.

1. Turn Off the Computer: Shut down the computer, switch off the power supply, and unplug the power cable.

2. Reseat the RAM: Open the RAM slot clips and carefully remove the RAM sticks. Reinsert them by aligning the notch with the slot and pressing until the clips lock.

3. Check RAM Slots: Install the RAM sticks in the recommended slots according to the motherboard manual. For two sticks, the recommended configuration is often A2 and B2.

4. Check RAM Compatibility: Ensure that the RAM type, capacity, and supported speed are compatible with the motherboard and CPU.

5. Reconfigure BIOS/UEFI: If the computer still fails POST, reset BIOS settings to their defaults. Disable XMP/EXPO temporarily so the RAM runs at a supported default speed.

6. Test Each RAM Stick: Start the computer with one RAM stick at a time to identify any faulty or incompatible module.

7. Restart and Verify: Turn on the computer, enter BIOS/UEFI, and confirm that the correct RAM capacity is detected and the system passes POST.

- ❖ Confirm the system boots normally after the fix.

1. Power On: Switch on the computer after reseating the RAM and correcting the BIOS/UEFI settings.

2. Check POST: Confirm that the computer completes the Power-On Self-Test without any RAM errors or warning beeps.

3. Check BIOS/UEFI: Verify that the correct RAM capacity and storage device are detected.

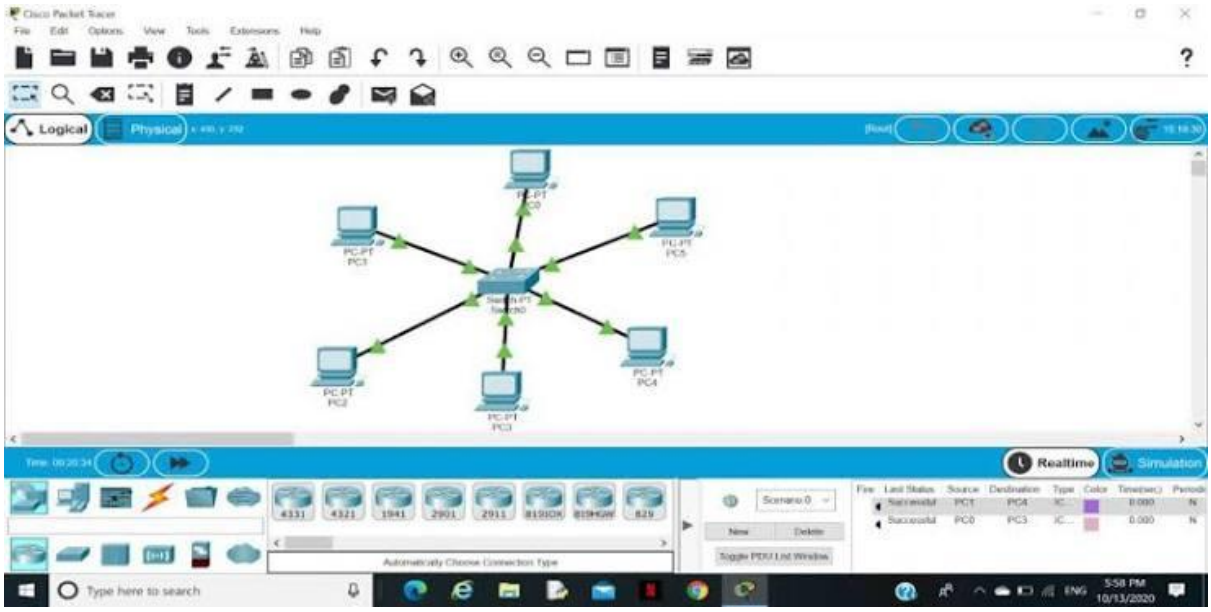
4. Check Operating System: Confirm that the computer loads Windows or the installed operating system normally.

5. Test System Stability: Open a few applications and check that the computer runs without freezing or restarting unexpectedly.

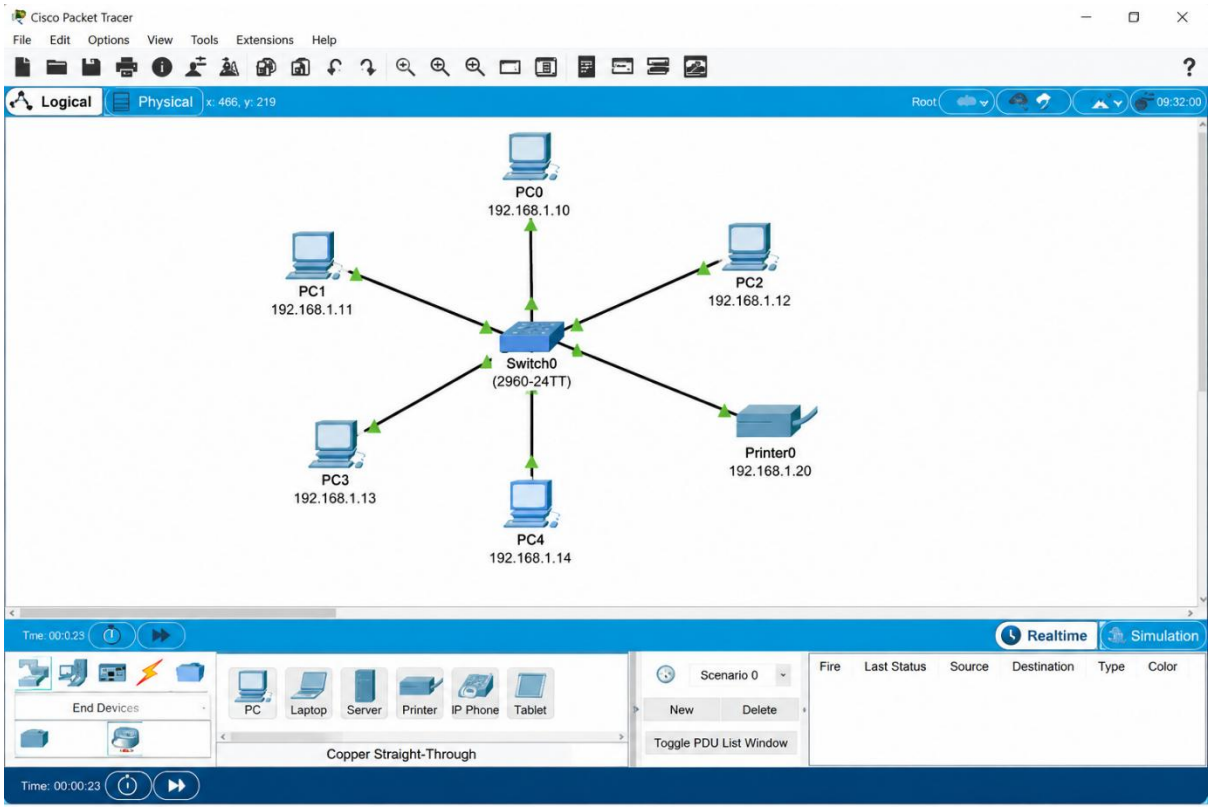
- Task 3:

LAN Topology Build

- Use Cisco Packet Tracer to build the topology.



❖ Include 5 end-systems and 1 network printer.



- Assign IP addresses from a Class C subnet to all devices.

Use the Class C network 192.168.1.0/24 for all five PCs and the network printer.

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
PC2	192.168.1.12	255.255.255.0
PC3	192.168.1.13	255.255.255.0
PC4	192.168.1.14	255.255.255.0
Printer0	192.168.1.20	255.255.255.0

- ❖ All assigned IPs must be reachable via Ping.

From example

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

ping 192.168.1.20

Test result	**Meaning**
Reply from 192.168.1.x	Ping successful
Request timed out	Device may be unreachable
Destination host unreachable	Check the IP configuration and network connection

- Task 4:

Clean OS Installation

- Perform a clean install of Windows/Linux from ISO.

Steps to Install the Operating System

1. Download the ISO: Download the Windows or Linux ISO file from its official website.

2. Create a Bootable USB: Use a tool such as Rufus to copy the ISO to a USB drive (usually 8 GB or larger).

3. Boot from USB: Insert the USB drive, restart the computer, and open the Boot Menu using the manufacturer's key, often F12, F9, Esc, or Del.

4. Start Installation: Select the USB drive and follow the installation instructions on the screen.

5. Choose Installation Type: Select the clean installation option. For Windows, choose Custom: Install Windows only when offered.

6. Select the Drive: Choose the intended installation drive or partition. Formatting or deleting partitions permanently removes their data, so confirm backups and the correct drive before proceeding.

7. Install the OS: Follow the prompts and wait while the installer copies files and restarts the computer.

8. Complete Initial Setup: Set the language, region, user account, password, and network connection.

9. Install Updates and Drivers: Install operating system updates and any missing drivers.

10. Verify the Installation: Restart the computer and confirm that the operating system boots normally and detects the CPU, RAM, storage, and network adapter.

- ❖ Manually partition the disk (not guided/automatic partitioning).

Steps

1. Boot from the ISO: Insert the bootable USB and start the computer from it.

2. Start Installation: Choose the language and keyboard settings, then begin the operating system installation.

3. Select Manual Partitioning:

- Windows: Select Custom: Install Windows only (advanced).
- Linux: Select Manual partitioning or Something else, depending on the installer.

4. Select the Disk: Identify the correct SSD or HDD. Check the disk size carefully before making changes.

5. Create Partitions: For a sample 100 GB disk, you could use the following layout for Linux:

Partition	Size	File system / Type	Mount point
EFI System Partition	512 MB	FAT32	/boot/efi
Root partition	40 GB	ext4	/
Home partition	58 GB	ext4	/home
Swap	1.5 GB	Swap area	—

These are example sizes; the exact layout depends on the operating system and available disk space. The partition sizes must fit within the actual disk capacity.

6. Assign the Mount Points: Set each partition's mount point as shown in the table. For Linux, use the EFI partition already present when appropriate on a UEFI system.

7. Confirm the Changes: Review the partition layout before applying it. Formatting or deleting a partition can permanently erase its data.

8. Install the OS: Select the root partition (/) for Linux or the intended Windows installation partition, then continue with installation.

9. Restart the Computer: After installation completes, remove the USB when prompted and restart the system.

10. Verify the Partitions: Open Disk Management in Windows or run lsblk in Linux to check that the partitions were created correctly.

- ❖ Optimize startup programs/services for faster boot.

Steps

1. Open Task Manager: Press Ctrl + Shift + Esc to open Task Manager.

2. Check Startup Apps: Select the Startup apps tab in Windows 11, or the Startup tab in older Windows versions.

3. Disable Unnecessary Apps: Right-click unnecessary applications, such as game launchers or apps that do not need to start automatically, and select Disable.

4. Keep Essential Apps Enabled: Do not disable antivirus software, security tools, or essential hardware and touchpad utilities without a clear reason.

5. Review Background Services: Press Win + R, type services.msc, and press Enter. Review services carefully; do not disable unfamiliar Windows or driver services.

6. Restart the Computer: Restart the system to apply the changes.

7. Check Boot Performance: Observe how long the computer takes to start and confirm that essential applications and hardware work correctly.

- ❖ OS must boot without errors after installation.

Verify That the OS Boots Without Errors

1. Restart the computer: After installing Windows or Linux, restart the system.
2. Check the boot process: Make sure the operating system loads without error messages or a boot loop.
3. Check device drivers: Open Device Manager in Windows or check hardware status in Linux to identify missing drivers.
4. Test basic functions: Check the keyboard, mouse, display, internet connection, and storage.
5. Install updates: Install the required OS updates and restart the computer again.
6. Verify stability: Make sure the system starts normally and works without freezing or crashing.

Section C

Mini Capstone Project

- Mini Project: Nexus Innovations — Diagnosing Random System Reboots

- Objective:

At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

- project must:

Identify the three hardware components most likely to cause random reboots under load.

- Explain why the systems pass POST in the morning but fail in the afternoon.

1. Overheating: The CPU or GPU may become too hot after hours of use.
2. Dust buildup: Dust can block fans and air vents, reducing cooling.
3. Faulty RAM: Defective or poorly seated RAM may cause intermittent boot failures.
4. Power supply problems: A weak or failing PSU may not provide stable power when the system is under load.
5. Loose connections: Heat expansion can sometimes affect loose cables or components.

Troubleshooting: Check CPU temperatures, clean fans and vents, test the RAM, check PSU stability, and inspect internal connections.

❖ Describe how you would use a stress-test tool to confirm a thermal or power issue.

 Install monitoring software: Use a tool such as HWMonitor to monitor CPU/GPU temperatures and power readings.

 Run a stress test: Use a tool such as OCCT to place a controlled load on the CPU or GPU.

 Monitor temperatures: Check whether temperatures rise too high or the CPU/GPU reduces performance due to overheating.

 Check for power problems: Watch for sudden shutdowns, restarts, crashes, or errors under load. These may indicate a PSU or power-delivery issue.

 Stop the test if necessary: Stop immediately if temperatures reach unsafe limits or the system becomes unstable.

 Record the results: Compare temperatures and system behaviour before, during, and after the test.

- ❖ Propose three specific steps to remediate the frequent system restarts.

1. Fix overheating: Clean dust from fans and air vents, check that the CPU cooler works properly, and monitor temperatures using HWMonitor.

2. Check the power supply (PSU): Inspect power cables and connections. Test the PSU with suitable equipment or replace it with a known-good, compatible unit if it is faulty.

3. Test RAM and update drivers: Run Windows Memory Diagnostic to check RAM errors and update the required hardware drivers. Reseat or replace faulty RAM if necessary.

- ❖ Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

Technical Justification for a PSU Upgrade

A PSU (Power Supply Unit) upgrade may be necessary if the current PSU cannot supply enough stable power to the CPU, GPU, and other components. An undersized or failing PSU can cause unexpected restarts, crashes, or shutdowns under heavy workloads.

However, an upgrade is not necessary if the PSU provides sufficient, stable power and is working correctly. First, check the PSU's capacity, connectors, and condition, and test it if possible.

Section D

AI-Augmented Learning

- **Step 1: AI-generated subnetting plan**

1. Exact prompt used

The following prompt can be entered into ChatGPT or another AI tool. It is written as the exact prompt for this assignment.

2. AI's original subnetting plan

For this example, the AI selects the private network 192.168.1.0/24.

Network subnet

192.168.1.0/24

Subnet mask

255.255.255.0

Usable host range

192.168.1.1 – 192.168.1.254

Broadcast address

192.168.1.255

Total usable host addresses

254

Original IP address allocation

Device	IP address	Subnet mask
System 1 (PC1)	192.168.1.10	255.255.255.0
System 2 (PC2)	192.168.1.11	255.255.255.0
System 3 (PC3)	192.168.1.12	255.255.255.0
System 4 (PC4)	192.168.1.13	255.255.255.0
System 5 (PC5)	192.168.1.14	255.255.255.0
Network Printer	192.168.1.20	255.255.255.0

Original reservation: Addresses 192.168.1.21 to 192.168.1.50 are reserved for future computers, printers, or other devices.

- **Step 2: Test and debug without AI**

1. Manual verification

I checked the original plan manually to identify subnetting errors.

\text{Learn more}

Verification check	Result	Explanation
Network address	Correct	192.168.1.0 is the network address.
Subnet mask	Correct	/24 equals 255.255.255.0.
Usable host range	Correct	.1 to .254 are usable host addresses.
Broadcast address	Correct	192.168.1.255 is the broadcast address.
Host capacity	Sufficient	254 usable addresses support all 6 devices.
Future expansion	Valid	.21 to .50 are available for future devices if kept unassigned.
Address overlap	None	All assigned addresses are unique and inside the subnet.

2. Corrected subnetting plan

The manual check found no mathematical subnetting errors in the original plan. The corrected version below keeps the same subnet and makes the address reservations clearer.

Final IP address allocation

Device	IP address	Purpose
System 1 (PC1)	192.168.1.10	Employee computer
System 2 (PC2)	192.168.1.11	Employee computer
System 3 (PC3)	192.168.1.12	Employee computer
System 4 (PC4)	192.168.1.13	Employee computer
System 5 (PC5)	192.168.1.14	Employee computer
Network Printer	192.168.1.20	Shared office printer
Future expansion	192.168.1.21–192.168.1.50	Reserved for new devices
Gateway (if needed)	192.168.1.1	Router address

Final subnet details

- Network address: 192.168.1.0
- CIDR prefix: /24
- Subnet mask: 255.255.255.0
- Usable host range: 192.168.1.1 – 192.168.1.254
- Broadcast address: 192.168.1.255
- Total usable addresses: 254
- Reserved for future expansion: 30 addresses (.21 to .50)

The gateway address .1 should be assigned to the actual office router if one is present. The reserved range must remain unassigned to other devices.